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Research on ChatGPT generated text detection model based on phonetic feature extraction and semantic features

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With the widespread adoption of large language models such as ChatGPT, distinguishing AI-generated text from human-written content has become increasingly challenging. Existing detection methods often rely solely on semantic representations and exhibit limited robustness, particularly when texts are paraphrased or rewritten. This study proposes an integrated detection framework that combines contextual semantic embeddings with auxiliary surface-level features, including pronunciation-related textual cues and handcrafted statistical descriptors. Specifically, a RoBERTa encoder is employed to capture deep contextual semantics, while a convolutional neural network aggregates multi-scale representations. In parallel, a set of text-derived structural, lexical, and readability features—serving as proxies for phonetic and stylistic regularities—are incorporated to enrich the representation space. Rather than introducing a fundamentally new detection paradigm, the proposed approach emphasizes feature-level fusion and systematic empirical evaluation. Experiments on the HAGTC dataset and a ChatGPT-written abstract dataset show that the proposed RoBERTa–CNN framework consistently outperforms several strong baselines in terms of accuracy and F1 score. Notably, the model demonstrates improved robustness in detecting rewritten AI-generated texts. Ablation studies further confirm that integrating multiple feature types significantly enhances detection performance. These results indicate that combining contextual representations with auxiliary surface features offers a practical and effective direction for AI-generated text detection.

Keywords Phonetic feature extraction, Semantic features, ChatGPT, Generate a text detection model

With the rapid development of large language models (LLMs) such as ChatGPT, AI-generated text has become increasingly fluent and contextually coherent, often approaching the quality of human writing[1–4]. This evolution has brought substantial benefits to content creation, education, and information services[5,6]. At the same time, it poses serious challenges to information authenticity and credibility in domains such as academic publishing, online media, and education[7–10]. As AI-generated content becomes progressively more difficult to distinguish from human-authored text, the ability to reliably identify its origin has emerged as an urgent and critical research problem.

Existing approaches to AI-generated text detection primarily rely on semantic representations learned by neural classifiers. Models based on pretrained language encoders, such as BERT and RoBERTa, have demonstrated strong performance on directly generated content[11–13]. However, recent studies have shown that their effectiveness degrades significantly when texts are paraphrased, lightly edited, or rewritten[2,14,15]. This limitation reveals a critical research gap: current methods rely heavily on deep semantic representations while largely overlooking surface-level regularities that remain relatively stable under text transformations. As a result, purely semantic modeling is often insufficient to capture subtle stylistic and structural patterns that distinguish machine-generated text from human writing, particularly under adversarial or post-editing scenarios[16,17].

To address this limitation, we investigate whether incorporating auxiliary surface-level cues beyond deep semantic embeddings can enhance detection robustness. In particular, we introduce a class of pronunciation-

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related textual proxies that approximate phonetic and prosodic regularities at the grapheme level, together with structural, lexical, and readability descriptors derived directly from text. Unlike conventional phonetic analysis based on speech signals, these proxies are extracted solely from textual patterns, capturing surface characteristics such as rhythm, repetition, and stylistic uniformity. By complementing contextual embeddings with these auxiliary cues, the representation space for detection can be effectively enriched.

Based on this motivation, we propose an integrated detection framework that combines a RoBERTa encoder with a convolutional neural network (CNN) and auxiliary text-derived features. RoBERTa is employed to capture deep contextual semantics, while the CNN aggregates multi-scale representations from hidden states. In parallel, handcrafted surface-level features—covering structural, lexical, and readability aspects—are incorporated to provide complementary information that is less sensitive to paraphrasing. Rather than introducing a fundamentally new detection paradigm, this study emphasizes a practical approach based on feature-level fusion and systematic empirical evaluation under realistic rewriting conditions.

The main contributions of this study are summarized as follows:

(1) We propose a feature-level fusion framework that integrates contextual embeddings from a pretrained language model with auxiliary surface-level textual features, enabling a more comprehensive representation for AI-generated text detection.
(2) We introduce pronunciation-related textual proxies as a novel category of surface-level features, which capture stylometric regularities that remain relatively stable under paraphrasing and rewriting.
(3) Through comprehensive experiments and ablation studies on two benchmark datasets, we demonstrate that feature fusion significantly improves detection robustness, particularly for rewritten or paraphrased AI-generated texts, and provide a systematic empirical analysis of its effectiveness.

Related work

With the rapid growth of social media and user-generated content, the misuse of large language models has raised increasing concerns regarding privacy leakage, misinformation, and academic integrity. Recent studies have shown that content generated by ChatGPT may expose sensitive information or exhibit stylistic patterns that differ from human writing, making reliable detection a critical component of content moderation systems[18]. In scientific writing, large language models often produce fluent but shallow text, which may lack domain-specific rigor and factual precision. Deep learning–based detectors have demonstrated the ability to capture subtle inconsistencies and stylistic deviations in such content, providing effective tools for identifying machine-generated academic text[6].

In professional domains such as medicine, the risks associated with machine-generated text are even more pronounced. Although large language models can rapidly produce plausible clinical narratives, these texts may contain oversimplifications, missing context, or factual inaccuracies. Prior work suggests that detection methods should go beyond grammatical fluency and incorporate indicators of knowledge consistency and professional style, particularly in high-stakes domains[19,20]. Similarly, analyses of ChatGPT-generated scientific abstracts reveal that, despite linguistic coherence, such texts often oversimplify complex concepts and omit critical background, leading to detectable deviations in depth and structure[21].

Beyond academic writing, generative models have been widely adopted in patent analysis, education, sentiment analysis, recommendation systems, and language learning[22–26]. While these applications demonstrate the practical value of generative AI, they also amplify the need for reliable detection mechanisms capable of distinguishing human-authored content from machine-generated text across diverse contexts.

Motivated by these challenges, this study investigates a practical detection framework for ChatGPT-generated text based on the fusion of contextual semantic representations and auxiliary surface-level cues derived directly from written text. Rather than relying on real speech signals, the proposed method interprets "phonetic" information as pronunciation-related textual regularities, such as rhythm, repetition, and boundary patterns at the character and token level. By collecting ChatGPT-generated and human-written texts, we construct benchmark datasets and analyze grammatical, semantic, and stylistic differences between the two sources. A multi-level deep learning model is then designed to integrate (i) contextual embeddings from a pretrained language model, (ii) multi-scale convolutional representations, and (iii) auxiliary surface-level features. This fusion-based architecture enhances robustness to paraphrasing and rewriting while maintaining computational efficiency. Extensive experiments are conducted to evaluate detection accuracy and efficiency in comparison with existing methods, aiming to provide a practical and reproducible solution for AI-generated text detection.

Method

Model analysis

In the proposed ChatGPT-generated text detection framework, the overall architecture consists of five components: text input, semantic encoding, surface-level feature extraction, multi-scale convolutional aggregation, and classification. Given an input document, a RoBERTa encoder is first applied to generate contextual embeddings that capture deep semantic information. In parallel, a set of auxiliary surface-level features—including pronunciation-related textual cues and handcrafted structural, lexical, and readability descriptors—are extracted directly from the raw text.

The hidden states produced by RoBERTa are further processed by adaptive pooling and convolutional layers to obtain multi-scale representations that summarize both local and global patterns. The resulting convolutional features are concatenated with the [CLS] embedding and the auxiliary surface-level feature vectors to form a unified representation. This fused representation is finally passed to a fully connected classification layer to predict whether the input text is human-written or AI-generated. The overall processing flow is illustrated in Fig. 1.

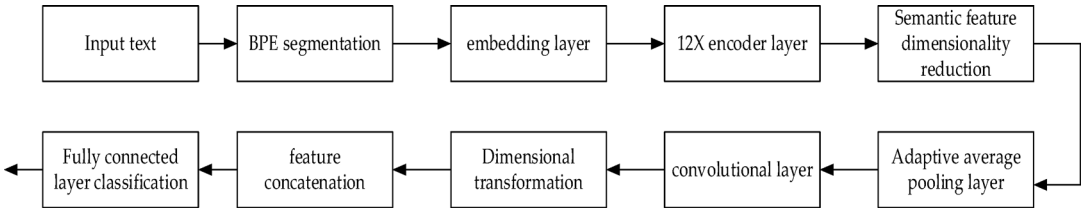


Figure 1. Text processing flow chart based on convolution method.

Given an input document, the text is first tokenized and fed into the RoBERTa encoder, which produces a sequence of contextualized hidden states

$$H = \{h_1, h_2, \ldots, h_L\}, \quad h_i \in \mathbb{R}^d,$$

where L denotes the sequence length and d is the hidden dimension. These representations capture rich contextual and semantic information for each token. The hidden state corresponding to the special [CLS] token, denoted as h_{cls}, is used as a global semantic summary of the entire input sequence.

In parallel, the full hidden-state matrix H is processed by an adaptive average pooling layer to reduce its temporal dimension and obtain a compact representation of the sequence. The pooled features are then passed through convolutional layers with different kernel sizes to extract multi-scale patterns that summarize both local and global characteristics of the text. This process yields a convolutional feature vector $h_{cnn} \in R^{dc}$.

The final representation of a document is constructed by concatenating the CLS embedding, the convolutional features, and the auxiliary surface-level feature vector fff derived from text:

$$h = [h_{\text{cls}}; h_{\text{cnn}}; f].$$

This fused vector integrates deep contextual semantics with complementary surface-level cues. It is subsequently fed into a fully connected layer followed by a softmax function to perform binary classification between human-written and AI-generated text. The entire model is trained end-to-end using the cross-entropy loss.

During training, the parameters of RoBERTa are fine-tuned together with the convolutional and classification layers. This joint optimization enables the model to adapt semantic representations to the detection task while preserving the benefits of pretrained knowledge. By combining contextual embeddings with multi-scale convolutional features and auxiliary surface-level descriptors, the proposed framework enhances detection robustness while maintaining computational efficiency.

Text detection methods

(1) Pronunciation-Related Textual Cues

In this study, we design a set of pronunciation-related textual cues that are derived directly from written text and serve as proxies for phonetic and prosodic regularities at the grapheme level. These cues aim to capture surface patterns such as rhythm, repetition, and stylistic uniformity, which may systematically differ between AI-generated and human-written text.

Specifically, we extract statistics from character and token sequences that reflect pronunciation-related regularities, including: (i) distributions of word and character lengths, (ii) frequencies of repeated n-grams and character patterns, (iii) punctuation density and interval patterns that approximate pause and rhythm structures, and (iv) token-level repetition rates and boundary regularities. Although these features are computed purely from text, they approximate properties associated with pronunciation and prosody, such as uniform rhythm, limited variation, and mechanical repetition, which are often observed in machine-generated content.

Each document is mapped to a fixed-length vector

$$p \in \mathbb{R}^{d_p},$$

where d_p denotes the dimensionality of the pronunciation-related feature space. These vectors are normalized and concatenated with other surface-level descriptors and semantic representations in subsequent stages of the model. By introducing these text-derived proxies, the framework incorporates complementary information that is less sensitive to paraphrasing and semantic rewording, thereby improving robustness in detecting rewritten AI-generated texts.

This design allows the model to exploit surface-level regularities without relying on real speech signals or text-to-speech pipelines, ensuring consistency between the conceptual formulation and the actual experimental implementation.

(2) Semantic extraction

In this study, RoBERTa model is used as the main semantic extraction tool, and BPE(Byte Pair Encoding) algorithm is combined to preprocess the text to optimize the effect of semantic representation. RoBERTa model is shown in the following Fig. 2:

In this study, RoBERTa is adopted as the primary semantic representation model. Input documents are tokenized using the standard Byte Pair Encoding (BPE) scheme employed by RoBERTa. The tokenized sequences are converted into input IDs and attention masks, which are subsequently fed into the encoder.

The embedding layer maps discrete tokens into continuous vector representations while incorporating positional information, forming the input to the Transformer encoder stack. RoBERTa employs a multi-layer Transformer architecture to capture contextual dependencies among tokens. The detailed mechanism of self-attention is omitted here for brevity, as it follows the standard Transformer formulation.

After passing through multiple encoder layers, RoBERTa produces deep contextualized representations. The hidden state corresponding to the [CLS] token is extracted as the global semantic representation of the input text and used for subsequent feature fusion and classification.

(3) Semantic classification

The core goal of semantic classification is to extract the global semantic features of text and classify it efficiently. In this study, RoBERTa model and convolutional neural network (CNN) are combined to achieve this goal. The specific method includes two aspects: on the one hand, the CLS vector output by RoBERTa model is directly used, on the other hand, the global semantic information is introduced into the convolution module, and different levels of text features are extracted by convolution operation. The process of classification through global semantics is shown in Fig. 3.

RoBERTa model, after text processing, outputs the hidden state vector through the coding layer, in which CLS vector carries the deep semantic information of the whole text. This vector is directly extracted and used for subsequent text classification tasks. In addition, in order to further enhance the semantic features of the text, this study also uses convolution module to capture the global semantic information. The input of convolution module includes hidden layer vector extracted by RoBERTa model, and different granularity features of text are extracted by convolution operation.

The convolutional module is designed to aggregate multi-scale representations from the RoBERTa hidden states and consists of three adaptive average pooling layers, two convolutional layers, and a fully connected layer. The pooling operations aim to reduce computational complexity while preserving salient semantic information. Specifically, the hidden-state matrix produced by RoBERTa is first processed by an adaptive average pooling layer that compresses its width to one-third of the original size. To further align feature dimensions and stabilize subsequent convolution operations, two additional pooling steps are applied: the first performs uniform pooling over the intermediate representation, and the second halves the sequence dimension, reducing the hidden-state resolution to compact forms with lengths of 2 and 3, respectively. This hierarchical compression ensures that the features extracted by different convolution kernels remain dimensionally compatible and comparable.

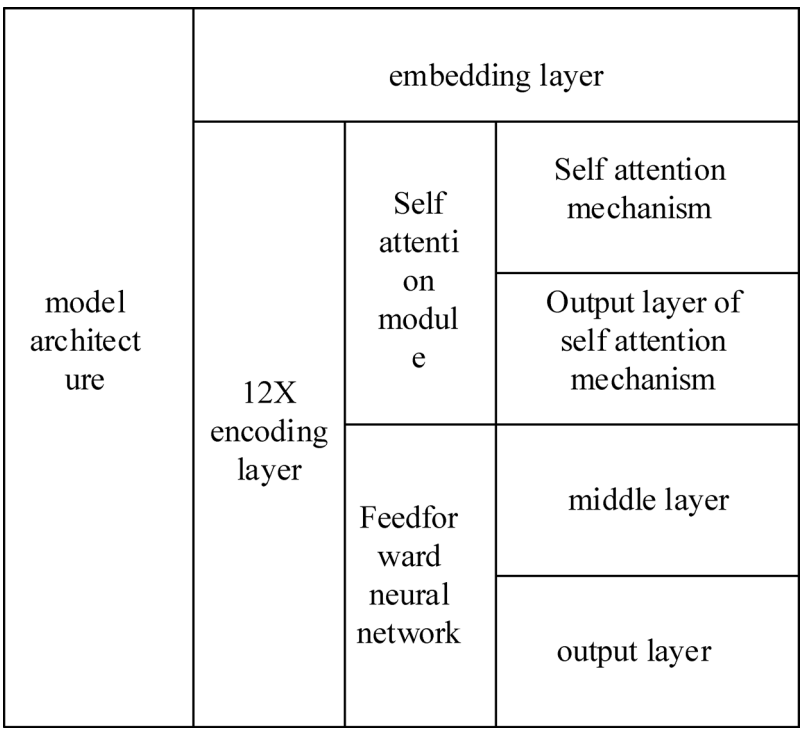


Figure 2. Schematic diagram of Roberta model.

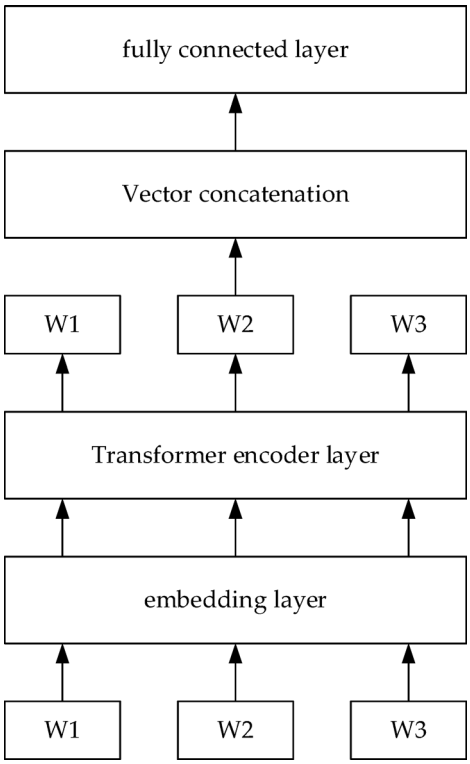


Figure 3. Schematic diagram of classification.

Feature type	Feature description
Architectural feature	Personal pronouns, punctuation marks, special symbols, etc
Lexical features	Paragraph sentence length, sentence unique words, number of spaces, etc
Language feature	Reading difficulty, complex vocabulary, readability, emotional factors, etc

Table 1. Text features.

In the convolution stage, kernels with different receptive fields are employed to capture multi-scale textual patterns. Convolution filters of sizes $2 \times 22 \times 22 \times 2$ and $3 \times 33 \times 33 \times 3$ are applied to the pooled representations, enabling the model to extract both fine-grained local dependencies and broader contextual structures. The outputs of these convolutional operations are flattened and concatenated to form a comprehensive feature vector. This vector is then fused with the global semantic representation provided by the RoBERTa [CLS] embedding, yielding a high-dimensional representation that integrates global semantics with local textual details. The fused vector is subsequently fed into a fully connected layer for final classification.

During training, the parameters of the RoBERTa encoder are not frozen, allowing end-to-end fine-tuning so that the semantic representations can be optimized for the detection task. Owing to its lightweight design and limited number of hyperparameters, the convolutional module converges rapidly and provides stable feature aggregation. By combining RoBERTa's deep contextual encoding with the multi-scale feature extraction capability of the convolutional network, the proposed model achieves efficient and accurate performance in detecting ChatGPT-generated text.

(4) Feature Fusion and Classification

In the detection model of ChatGPT generated text based on phonetic feature extraction and semantic features, the extraction of semantic features is used to distinguish ChatGPT generated text from human-written text. In this study, semantic features are divided into three categories: structural features, lexical features and linguistic diversity features, and these features are used to capture the differences of text styles. The specific features contained in the three feature categories are shown in Table 1.

The core contribution of the proposed framework lies in its explicit feature-level fusion strategy, which integrates deep contextual semantics, multi-scale convolutional representations, and auxiliary surface-level cues into a unified representation for detection.

Given an input document, the RoBERTa encoder produces a contextual hidden-state sequence:

$$H = \{h_1, h_2, \ldots h_L\}, \quad h_i \in \mathbb{R}^d.$$

The hidden state corresponding to the special [CLS] token is extracted as a global semantic summary of the entire text:

$$h_{\text{cls}} \in \mathbb{R}^d.$$

In parallel, the hidden-state matrix H is processed by adaptive average pooling followed by convolutional layers with multiple kernel sizes. This module aggregates both local and global patterns in the sequence and outputs a fixed-length vector:

$$h_{\text{cnn}} \in \mathbb{R}^{d_c}.$$

In addition, a set of auxiliary surface-level features—including pronunciation-related textual cues and other stylometric descriptors described in "Text detection methods" section—are extracted directly from the raw text and normalized into a fixed-length vector:

$$f \in \mathbb{R}^{d_f}.$$

These three components are concatenated to form the final fused representation:

$$h = [h_{\text{cls}}; h_{\text{cnn}}; f].$$

This vector explicitly combines deep semantic information, multi-scale convolutional features, and surface-level stylistic cues. The fused representation is then passed to a fully connected layer followed by a softmax function to perform binary classification:

$$\hat{y} = \text{Softmax}(Wh + b),$$

where W and b denote the trainable weight matrix and bias term, respectively, and $\hat{y}$ represents the predicted probability distribution over human-written and AI-generated classes.

The entire network, including RoBERTa, convolutional layers, and the classification head, is trained end-to-end using the cross-entropy loss. This fusion-based design allows the model to jointly exploit contextual semantics and complementary surface-level regularities, enabling more robust detection, particularly for paraphrased or rewritten AI-generated texts.

Structural features do not directly convey semantic information, but help to reflect the organization and format of the text. In this study, stop words, punctuation, discourse markers, personal pronouns, capital letters and part-of-speech tagging are selected. The text generated by ChatGPT usually has low stop words frequency and simple punctuation, while human texts often have more modifiers and emotional expressions. By analyzing these structural features, we can effectively distinguish the stylistic differences between generated texts and human texts.

Lexical features focus on the complexity and information density of the text. By calculating the number of words, sentence length and paragraph length of the text, this study reflects the richness of the text. ChatGPT tends to generate concise and direct texts, with low vocabulary diversity, while the texts written by humans pay more attention to details and modification, with high vocabulary change and complexity. By calculating the mean and variance of unique words and all words in the text, it is further revealed whether the text contains repeated or oversimplified information.

The characteristics of language diversity include readability of text, vocabulary richness, reading difficulty and emotional analysis. The text generated by ChatGPT is usually clear in logic, regular in structure and neutral in emotional expression, while the human text may be more complex, with more prominent emotional polarity and subjectivity. By analyzing the readability and emotional polarity of the text, we can further identify the source of the text.

Experiment

Experimental environment

The experimental environment is configured as follows: Ubuntu 20.04.4 as the operating system, Intel Xeon Gold 6240 as the processor, NVIDIA A100 as the graphics card and Linux 5.10.0-8-generic as the kernel version. The development platform uses Anaconda 3.7, the programming language is Python 3.7.9, the deep learning framework is PyTorch 1.10.0, and the program editor is VS Code. The model training procedure follows standard fine-tuning settings commonly adopted for pretrained language models. The learning rate was set to 2×10^{-5}, and optimization was performed using the AdamW optimizer. The batch size was fixed at 16, and the model was trained for 5 epochs. Early stopping based on validation performance was employed to prevent overfitting.

Data samples

In this study, the Human-and AI-Generated Texts Classification (HAGTC) is selected, which comes from the educational environment and covers ten different topics, and can effectively distinguish the characteristics of human-and AI-generated texts. ChatGPT is used to compile the ChatGPT-written Abstract Dataset. The text in this dataset involves computer science and covers the scientific research abstracts generated by Chatgpt. These data sets provide multi-angle text data support for model training and verification in this study. The

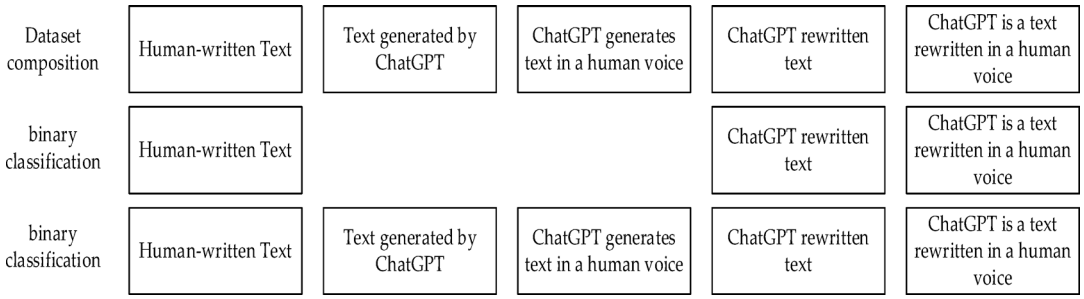


Figure 4. Schematic diagram of document partitioning in HAGTC dataset.

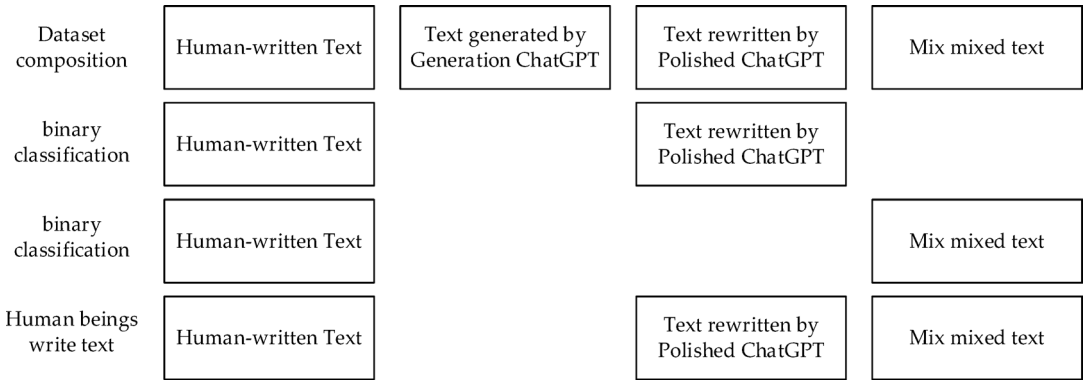


Figure 5. Schematic diagram of CHAET dataset partitioning.

HAGTC dataset contains some samples, evenly distributed between human-written and AI-generated texts. We randomly split the dataset into training (70%), validation (10%), and testing (20%) sets while preserving class balance. Rewritten texts were generated using ChatGPT with paraphrasing prompts to simulate realistic post-editing scenarios. To ensure reproducibility and statistical reliability, the dataset was randomly partitioned into training (70%), validation (10%), and testing (20%) subsets while preserving class balance across all splits. The partitioning procedure was conducted using stratified sampling, ensuring that both authorship categories and topic distributions remained consistent across subsets.

In addition to directly generated AI texts, we constructed a rewritten subset to evaluate model robustness under paraphrasing conditions. Specifically, AI-generated texts were reprocessed using ChatGPT with controlled paraphrasing prompts designed to simulate realistic post-editing scenarios. The rewriting prompts instructed the model to preserve the original semantic content while altering lexical choice, sentence structure, and syntactic organization. This strategy aims to emulate adversarial or human-assisted rewriting conditions commonly observed in real-world applications.The rewritten texts were generated under consistent decoding settings to minimize variability introduced by sampling randomness. Both the original and rewritten subsets were incorporated into the evaluation pipeline to assess detection performance under varying degrees of textual modification.

These data sets provide multi-angle text data support for model training and verification in this study.

The document partitioning of the HAGTC dataset is shown in the following Fig. 4:

The CHAET dataset is divided as shown in the following Fig. 5:

Evaluation indicators

In the evaluation of model performance, this article mainly uses accuracy and F1 score for evaluation and analysis.

Accuracy refers to the proportion of samples correctly predicted by a model to the total number of samples. It is one of the most intuitive evaluation indicators, reflecting the overall predictive ability of the model across all categories. The formula is as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

In the above formula, TP refers to the number of samples correctly predicted as positive class, TN refers to the number of samples correctly predicted as negative class, FP refers to the number of samples incorrectly predicted as positive class, and FN refers to the number of samples incorrectly predicted as negative class. The accuracy can be divided into the following formula:

$$Accuracy = \frac{TP}{TP+FP} + \frac{TP+FP}{TP+TN+FP+FN} + \frac{TN}{TN+FN} + \frac{TN+FN}{TP+TN+FP+FN} \quad (2)$$

Accuracy refers to the proportion of samples predicted by the model as positive classes that are truly positive. The formula is as follows:

$$\Pr ecision = \frac{TP}{TP+FP} \quad (3)$$

The recall rate is the proportion of samples correctly predicted as positive by the model to all positive samples:

$$\text{Recall} = \frac{TP}{TP+FN} \quad (4)$$

The F1 value is the harmonic mean of precision and recall, taking into account both the accuracy and comprehensiveness of the model. The formula is as follows:

$$F1 = 2 * \frac{\Pr ecision * \text{Recall}}{\Pr ecision + \text{Recall}} \quad (5)$$

By substituting the precision and recall into formula (5), we can obtain:

$$F1 = 2 * \frac{\frac{TP}{TP+FP} * \frac{TP}{TP+FN}}{\frac{TP}{TP+FP} + \frac{TP}{TP+FN}} \quad (6)$$

Comparative study

To ensure statistical reliability, each experiment was repeated five times with different random seeds. We report the mean and standard deviation of Accuracy and F1 score. In addition, paired t-tests were conducted between the proposed model and the strongest baseline to assess statistical significance ($p < 0.05$).

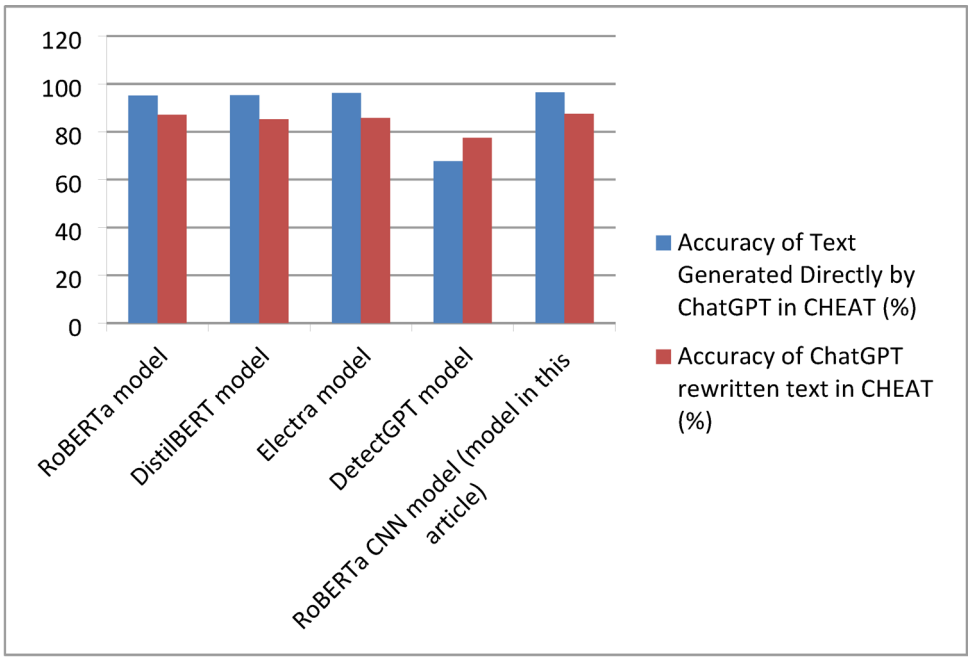
In order to reduce the influence of under-fitting on the experimental results, each model is experimented on this HAGTC set for many times, and the average of accuracy and F1 value is taken to better verify the detection effect of the model on ChatGPT generated text. And this part also selects RoBERTa model(Li et al.,2024), DistilBERT model(Venkatesh et al.,2022), Electra model(Zhu et al.,2023) and DetectGPT model(Aar et al.,2025) to compare with RoBERTa-CNN model in this study. See Table 2 below for the results of comparative experiments on HAGTC data sets:

As can be seen from Table 2, RoBERTa model achieved 87.71% accuracy and 92.42% F1 value on "the text directly generated by ChatGPT", while the accuracy was slightly lower on "the text rewritten by ChatGPT", with 83.22% and F1 value of 87.33%. In contrast, DistilBERT model improved slightly, with the accuracy of 87.74% and F1 value of 92.61%, but the accuracy of "ChatGPT rewritten text" dropped significantly to 80.15% and F1 value of 85.24%. This shows that although DistilBERT can provide high accuracy, it is not as robust as RoBERTa when dealing with ChatGPT rewritten text. Electra model is the most outstanding in "the text directly generated by ChatGPT", with an accuracy of 89.15% and a F1 value of 93.13%, but its performance in "the text rewritten by ChatGPT" has declined, with an accuracy of 79.28% and a F1 value of 83.69%. This shows that the Electra model has obvious advantages in the face of the original generated text, but its ability to process the rewritten text is relatively weak. DetectGPT model is the worst among all models, especially on "ChatGPT rewritten text", the accuracy is only 66.83%, and the F1 value is 80.47%. Although it performs well in F1 value, its low accuracy shows that it is not ideal in the detection of ChatGPT generated text, which may be because the model pays more attention to the semantic level of the text and ignores the overall understanding of the generated text. RoBERTa-CNN model (the model in this study) is the best in the comparative experiment, especially in the "text directly generated by ChatGPT", the accuracy rate is as high as 91.46%, and the F1 value is 94.66%. On the "text rewritten by ChatGPT", the accuracy is 87.64% and the F1 value is 90.21%. Compared with other models, RoBERTa-CNN model proposed in this study maintains a high level of accuracy and F1 value, especially when dealing with rewritten text, showing strong robustness.

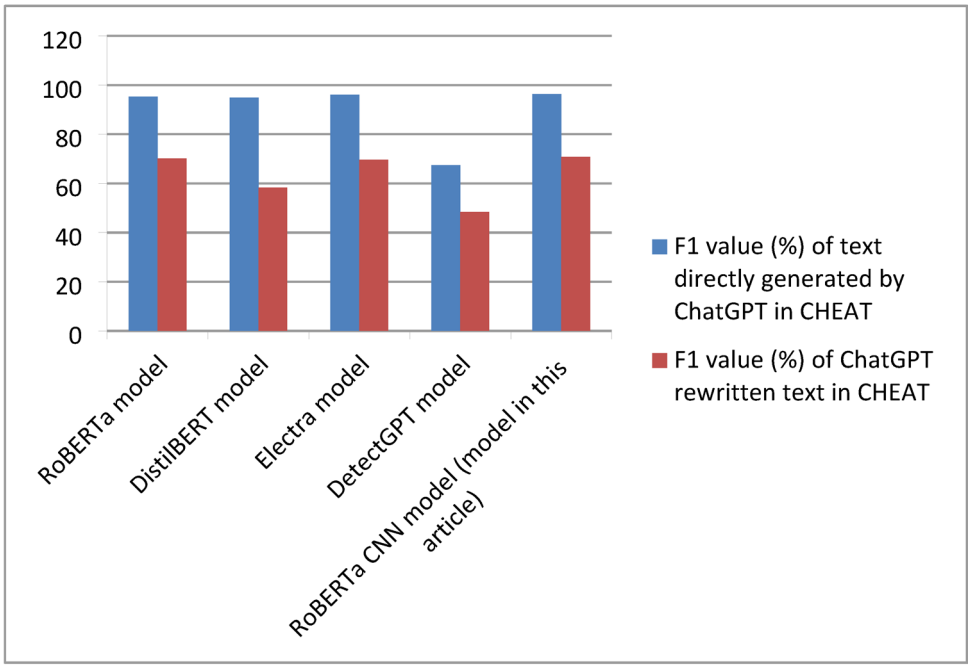
See Fig. 6 below for the results of comparative experiments on ChatGPT-written Abstract Dataset:

	Text directly generated by ChatGPT in HAGTC		Text rewritten by ChatGPT in HAGTC	
	Accuracy (%)	F1 value (%)	Accuracy (%)	F1 value (%)
RoBERTa model	87.71	92.42	83.22	87.33
DistilBERT model	87.74	92.61	80.15	85.24
Electra model	89.15	93.13	79.28	83.69
DetectGPT model	81.63	93.34	66.83	80.47
RoBERTa-CNN model (this model)	91.46 ± 0.32	94.66	87.64	90.21

Table 2. Comparative experiments on HAGTC data sets. The improvements of RoBERTa–CNN over RoBERTa are statistically significant ($p < 0.05$).



(a) Accuracy rate



(b) F1 value

Figure 6. Comparative experiment on HEAT dataset.

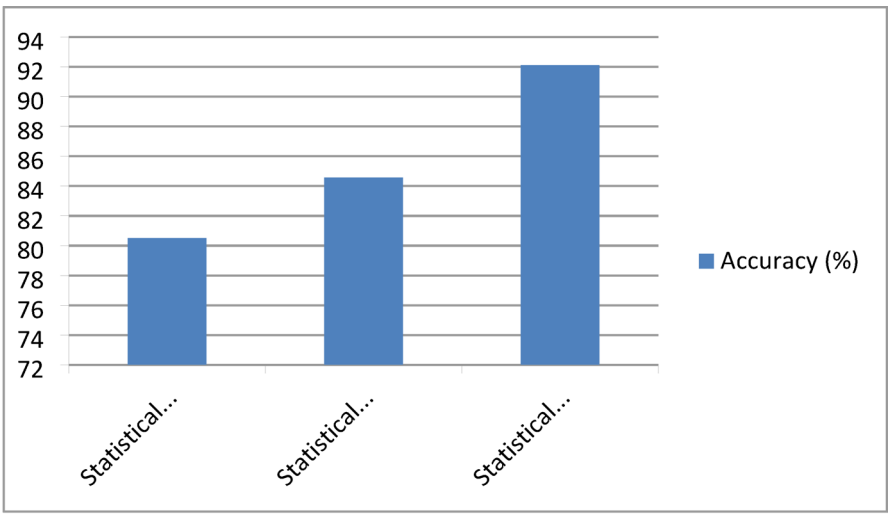
RoBERTa model performs well in "the text directly generated by ChatGPT", with an accuracy rate of 95.16% and a F1 value of 95.22%, but in "the text rewritten by ChatGPT", the accuracy rate drops to 87.11% and the F1 value is 70.13%. This result shows that RoBERTa model can achieve high detection accuracy for the original generated text, but its performance drops obviously when faced with rewritten text. DistilBERT model is similar to RoBERTa, with the accuracy of 95.25% and F1 value of 94.82% for "the text directly generated by ChatGPT", which is slightly higher than RoBERTa, but the accuracy of 85.26% and F1 value of "the text rewritten by ChatGPT" is 58.26%, which also shows a great performance decline. This shows that although DistilBERT can provide high accuracy in the original generated text, its robustness in dealing with rewritten text is weak. Electra model performs best in "the text directly generated by ChatGPT", with an accuracy rate of 96.17% and a F1 value of 96.11%, which is slightly higher than RoBERTa and DistilBERT. However, in "the text rewritten by ChatGPT", its accuracy rate and F1 value are 85.72% and 69.58%, respectively, and there is still a

certain degree of decline, indicating that its adaptability to rewriting text is limited. DetectGPT model is the worst among all the comparison models, especially on "the text directly generated by ChatGPT", the accuracy is only 67.72%, and the F1 value is 67.34%, and the performance on "the text rewritten by ChatGPT" is also weak, with the accuracy of 77.47% and the F1 value of 48.35%. This shows that DetectGPT model can't achieve the same efficient performance as other models when processing the text generated by ChatGPT. RoBERTa-CNN model (this model) shows the best comprehensive performance in comparative experiments. In "the text directly generated by ChatGPT", the accuracy rate is 96.43% and the F1 value is 96.37%; in "the text rewritten by ChatGPT", the accuracy rate is 87.53% and the F1 value is 70.84%. Compared with other models, RoBERTa-CNN model maintains a high accuracy, but also shows a strong advantage in the detection ability of rewritten text.

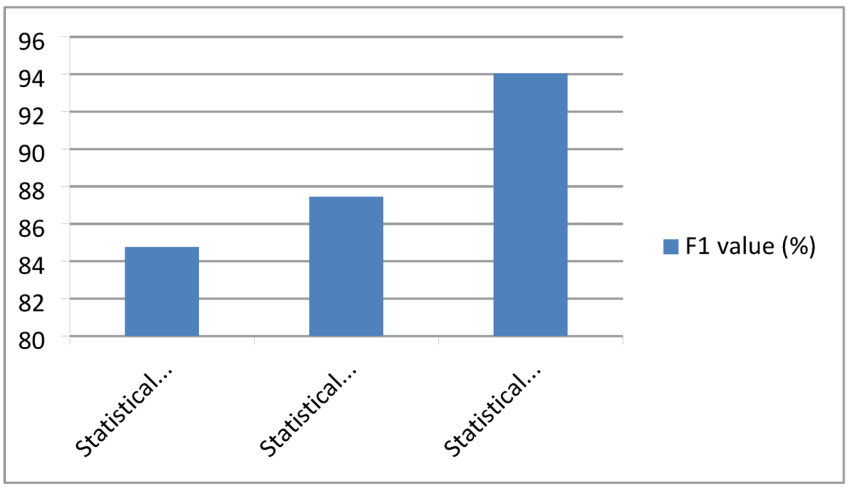
Ablation experiment

Aiming at the selection of text features, this part divides the features into three categories, namely statistical features, special tags and readability features, and carries out ablation experiments on paragraph texts of HAGTC data set. The experimental results of text feature ablation are shown in the following Fig. 7:

According to Fig. 7, the combination of statistical features and readability features is 80.51% and 84.76% in accuracy and F1 value. This combination mainly focuses on the basic structural information and readability of the text, but compared with other combinations, its accuracy and F1 value are relatively low. This shows that although these features can provide some basic information about text fluency and information density, they still have some limitations in detecting the subtle differences between ChatGPT-generated texts and human texts. The combination of statistical features and special marks has significantly improved the accuracy (84.56%) and F1 value (87.45%). Special tag features play an important role in capturing grammatical structure, punctuation and other semantic level information in the text, helping the model to better distinguish the text generated by



(a) Accuracy rate



(b) F1 value

Figure 7. Experimental results of text feature ablation.

Convolution kernel size	Accuracy (%)	F1 value (%)
2	78.77	79.27
Three	80.11	80.73
2, 3	81.10	81.35
2, 4	78.54	79.46
2, 3, 4	80.17	80.54

Table 3. Experimental results of convolution nuclear ablation.

ChatGPT from the text written by human beings. Compared with statistical features and readability features alone, the sensitivity and accuracy of the model in text processing are improved by adding special marking features. The combination of statistical features + readability features + special marks achieves 92.11% accuracy and 94.03% F1 value. This combination is significantly better than the other two combinations, which shows that the fusion of the three features can provide more comprehensive text information for the model, especially when combining the structure, readability and marking features of the text, it can capture the subtle differences in the text more accurately, thus improving the detection performance.

In this model, convolution and pooling operations are used to extract global semantic information, so ablation experiments are carried out for the selection of convolution kernel and model structure. Paragraph data with evenly distributed topics on HAGTC data set are adopted. The experimental results of convolution nuclear ablation are shown in the following Table 3:

According to Table 3, when the convolution kernel size is 2, the accuracy of the model is 78.77% and the F1 value is 79.27%. This result shows that although a single small convolution kernel can extract text features locally, its recognition ability is still weak, especially in the face of complex text structure, the capture of local features is limited, resulting in low accuracy and F1 value. When the convolution kernel size is 3, the accuracy is 80.11% and the F1 value is 80.73%. Compared with the case of convolution kernel size 2, the model with convolution kernel size 3 extends the scope of feature extraction and can capture more context information, thus slightly improving the detection performance. This shows that the larger convolution kernel can extract the global information of the text better, but its improvement is relatively limited. The combination of convolution kernel size 2 and 3 shows a significant improvement, with an accuracy of 81.10% and a F1 value of 81.35%. This combination can make use of the advantages of small convolution kernel and medium convolution kernel at the same time, and enhance the sensitivity to wider context information while extracting local features, thus improving the performance of the model. The performance of the combination with convolution kernel size of 2 and 4 decreased slightly, with the accuracy of 78.54% and the F1 value of 79.46%. Although a large convolution kernel is used, its feature range is too wide, which may lead to the loss of some detailed information, thus affecting the overall performance of the model. The performance of the combination with convolution kernel sizes of 2, 3 and 4 has rebounded, with an accuracy of 80.17% and a F1 value of 80.54%. This shows that when combining small, medium and large convolution kernels, the model can capture text features from multiple scales. Although the improvement is limited, it can find a balance point among features with different granularity, thus improving the overall performance of the model.

Error analysis

To better understand the limitations of the proposed detection framework, we conducted a qualitative analysis of representative misclassified samples. The observed error patterns reveal several systematic tendencies associated with both AI-generated and human-written texts.

We observe that misclassified AI-generated texts frequently contain **domain-specific terminology, technical expressions, and structurally complex sentences** that closely resemble formal academic writing. In such cases, the semantic representations extracted by RoBERTa capture coherent logical structures and specialized vocabulary distributions that are typically associated with human-authored scholarly texts. Moreover, rewritten AI-generated samples often exhibit increased syntactic variability and reduced repetition, thereby weakening surface-level cues that the model relies upon. As a result, these texts occasionally become indistinguishable from authentic academic prose.

Conversely, certain human-written texts are occasionally predicted as AI-generated. These instances commonly display **highly standardized formatting, low emotional variability, and reduced stylistic irregularities**. Texts characterized by concise sentence construction, uniform lexical selection, and neutral tone may inadvertently resemble machine-generated content. In particular, human-written academic abstracts and technical reports often emphasize clarity, consistency, and structural regularity, which can overlap with statistical patterns commonly observed in AI-generated texts.

These findings suggest that the detection task is fundamentally influenced by **stylistic regularization effects rather than purely authorship-dependent features**. When either AI-generated or human-written texts exhibit extreme stylistic normalization—such as highly uniform vocabulary, predictable syntactic structures, or suppressed expressive variation—the discriminative signals available to the classifier may be substantially reduced. Furthermore, the analysis indicates that rewriting operations primarily impact **surface-level statistical features**, including lexical diversity, repetition rates, and punctuation distributions, while preserving core semantic content. This observation explains why semantic-only detectors tend to experience performance degradation under paraphrasing conditions and highlights the importance of integrating heterogeneous feature types.

Overall, the misclassification patterns reflect an inherent challenge of AI-generated text detection: the increasing convergence of stylistic characteristics between advanced language model outputs and carefully edited human writing. Future work may address this limitation by incorporating discourse-level modeling, cross-document consistency analysis, and more fine-grained stylometric descriptors.

Conclusion

In this study, we address the problem of detecting ChatGPT-generated text by proposing a RoBERTa–CNN framework that integrates contextual semantic representations with auxiliary surface-level features. Rather than introducing a fundamentally new detection paradigm, the model emphasizes feature-level fusion: deep semantic embeddings produced by RoBERTa are combined with convolutionally aggregated representations and a set of text-derived structural, lexical, and readability descriptors, which serve as proxies for stylistic and pronunciation-related regularities observable in written text.

Experiments conducted on the HAGTC and CHEAT datasets show that this integrated framework achieves consistently higher accuracy and F1 scores than several representative baselines, with particularly improved stability on rewritten or paraphrased AI-generated texts. Feature ablation studies further indicate that no single component is sufficient on its own; instead, the joint use of contextual embeddings and auxiliary surface features contributes most to the observed performance gains.

The empirical results suggest that moderate yet consistent performance gains can be achieved through feature-level fusion, particularly under rewriting and paraphrasing conditions. Rather than relying on architectural novelty, the improvements observed in this study primarily stem from the complementary interaction between semantic representations and auxiliary surface-level cues.

These results suggest that practical improvements in AI-generated text detection can be achieved through careful fusion of heterogeneous feature types, even without relying on novel model architectures. Future work will focus on extending the framework to multilingual settings, refining the definition and extraction of surface-level features, and improving computational efficiency for large-scale deployment.

Data availability

The data sets generated and/or analyzed during this study cannot be disclosed due to the need for confidentiality, but can be obtained from the corresponding authors according to reasonable requirements.

Code availability

The custom code generated and/or used during this study not publicly available due to confidentiality requirements, but can be obtained from the corresponding authors upon reasonable request.

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Author contributions
All authors contributed significantly to this research. H.C. provided the ideas and experimental design, and HC.C. implemented the design plan, wrote the research paper, and performed proofreading. JH.C, BY.H, H.Z. , XT.W., JP.W.conducted data collection and analysis. All authors participated in manuscript preparation and proofreading, and approved the final version for submission. All authors have read and agreed to the published version of the manuscript.

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